Code No: 133AJ **R16**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B.Tech II Year I Semester Examinations, August/September - 2022
DIGITAL LOGIC DESIGN
(Common to CSE, IT)

Time: 3 Hours **Max.Marks:75**

Answer any five questions
All questions carry equal marks
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1.a) Convert the following numbers with the indicated bases to decimal:
i) $(12121)_3$ ii) $(198)_{12}$ iii) $(4310)_5$ iv) $(50)_7$
b) Using 2's complement perform $(42)_{10} - (68)_{10}$. [8+7]

2.a) What is Hamming code? How is the Hamming code word tested and corrected?
b) Perform the decimal addition of 679.6 + 536.8 in the 8421 code. [8+7]

3.a) Find the reduced POS form of the function .
F(A, B, C, D) = Σm(1, 3, 7, 11, 15) + Σd(0, 2, 5). Implement using NAND logic.
b) Explain about essential prime implicants with suitable example. [8+7]

4. Reduce the following functions using K-map techniques. [8+7]
a) f (A, B, C, D, E) = $\sum m(1, 4, 8, 10, 11, 20, 22, 24, 25, 26) + d(0, 12, 16, 17)$
b) f (A, B, C, D) = $\pi M(4, 5, 6, 7, 8, 12, 13) + d(1, 15)$.

5.a) What is multiplexer? Draw the logic diagram of 8 to 1 line multiplexer.
b) Design a gray to BCD code converter. [7+8]

6.a) Design half adder and full adder using logic gates with truth tables.
b) Design the 4 – bit binary adder with suitable logic diagram. [8+7]

7.a) Draw the structure of D Flip Flop and write truth table.
b) Compare combinational and sequential circuits.
c) Explain 4-bit shift register with parallel load with a neat diagram. [4+4+7]

8.a) Discuss about Programmable Array Logic (PAL).
b) Explain different types of memory. Explain Error detection and error correction of ROM. [7+8]

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